PHARMACY EXPIRY & RE-ORDER DISPATCH ENGINE

NAME: DEVARAJ M

SRN: PES1UG24AM079

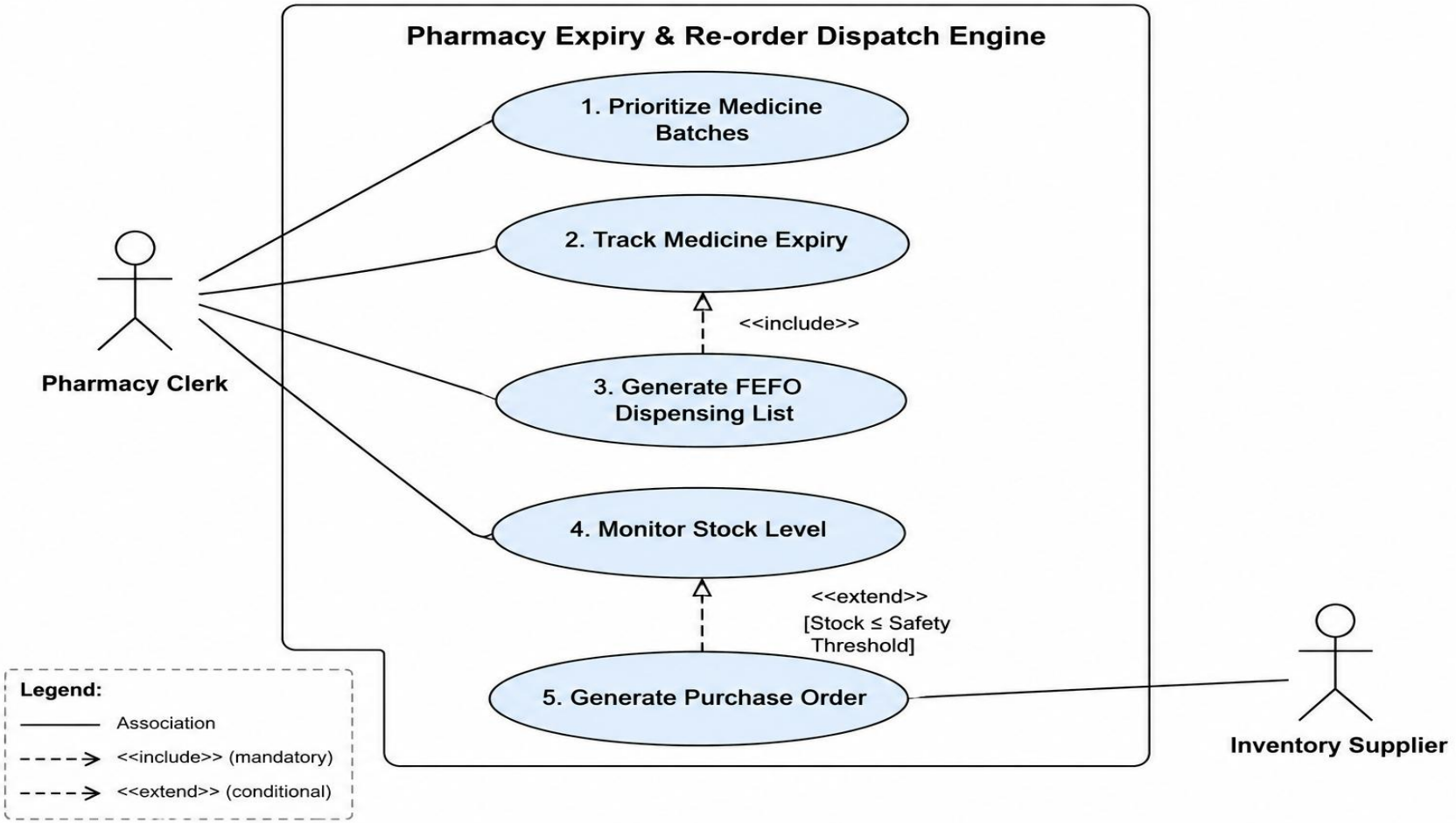
FUNCTIONAL REQUIREMENTS

ID	Type	Priority	Functional Requirement	Acceptance Criteria	Rationale
FR-001	Functional	High	The system shall enforce FEFO picking order by prioritizing medicine batches with the nearest expiry date during sales checkout.	**Pass:** Oldest valid batch is selected first. **Fail:** A newer batch is assigned while an older unexpired batch exists.	Ensures batches closest to expiry are used first and helps reduce medicine wastage.
FR-002	Functional	High	The system shall track the expiry dates of medicine batches stored in the pharmacy.	**Pass:** The system correctly records and displays the expiry date for each medicine batch. **Fail:** The system does not record or displays an incorrect expiry date.	Helps identify batches approaching expiry and prevents expired medicines from being dispensed.
FR-003	Functional	High	The system shall generate a FEFO dispensing list by arranging eligible medicine batches according to their expiry dates.	**Pass:** The dispensing list places the earliest-expiring valid batch first. **Fail:** A later-expiring batch is placed before an earlier valid batch.	Enables pharmacy staff to dispense medicines according to the FEFO principle.
FR-004	Functional	High	The system shall monitor medicine stock levels and compare the available quantity with the defined safety-stock threshold.	**Pass:** The system identifies when stock reaches or falls below the safety threshold. **Fail:** The system fails to identify stock at or below the threshold.	Helps the pharmacy identify medicines that require re-ordering.
FR-005	Functional	High	The system shall trigger an automated purchase-order request when medicine stock reaches or falls below the safety-stock threshold.	**Pass:** A purchase-order request is automatically triggered when the threshold is reached or crossed. **Fail:** No purchase-order request is triggered.	Ensures timely re-ordering and helps prevent medicine stock-outs.

NON FUNCTIONAL REQUIREMENTS

ID	Type	Priority	Non-Functional Requirement	Acceptance Criteria	Rationale
NFR-001	Performance	High	The system shall generate and dispatch supplier purchase-order requests within 2 seconds when medicine stock reaches or falls below the safety threshold.	Pass: Benchmarking under simulated peak load confirms that the purchase-order request is generated and dispatched within 2 seconds. Fail: The request takes more than 2 seconds.	Ensures timely re-ordering and reduces the risk of medicine stock-outs.
NFR-002	Security	High	The system shall ensure that only authorized pharmacy personnel can access inventory and purchase-order information.	Pass: Authorized personnel can access the information, while unauthorized users are denied access. Fail: An unauthorized user gains access.	Protects sensitive pharmacy and inventory information from unauthorized access.

UML DIAGRAM



USE CASE FLOW SPECIFICATION

Use Case: Generate FEFO Dispensing List
Actor: Pharmacy Clerk
Purpose: To generate a dispensing list that prioritizes medicine batches according to their expiry dates using the FEFO principle.

Preconditions

1. The Pharmacy Clerk is authenticated in the system.
2. Medicine batches are recorded in the system.
3. Each eligible medicine batch has a valid expiry date.
4. The pharmacy has available medicine stock.

Postconditions

1. A FEFO dispensing list is generated successfully.
2. Eligible medicine batches are arranged according to their expiry dates, with the earliest expiry first.
3. The generated list is displayed to the Pharmacy Clerk.

Main Success Scenario

1. The Pharmacy Clerk selects **"Generate FEFO Dispensing List."**
2. The system retrieves the available medicine batches.
3. The system retrieves the expiry dates of the available batches.
4. The system identifies eligible medicine batches for dispensing.
5. The system sorts the eligible batches by ascending expiry date.
6. The system generates the FEFO dispensing list.
7. The system displays the generated list to the Pharmacy Clerk.
8. The Pharmacy Clerk uses the list to dispense medicines in FEFO order.

Alternate Flow

A1. No eligible medicine batches are available

1. The system checks the available medicine batches.
2. The system finds that no eligible batches are available for dispensing.
3. The system displays **"No eligible medicine batches available for dispensing."**
4. The system does not generate a FEFO dispensing list.
5. The Pharmacy Clerk may update the inventory and try again.